HW2

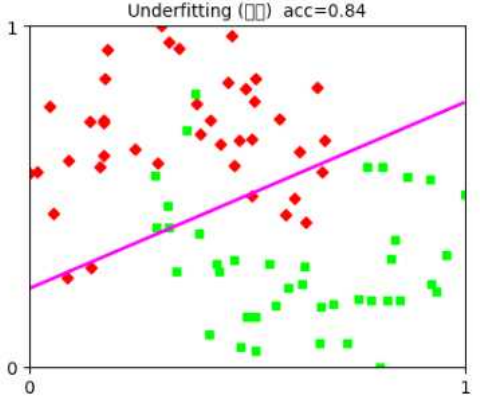
classification

20221219 이지원

간단한 모델

```
model = Sequential()
# first dense layer
model.add(Dense(units=2, input_shape=(2,), activation='sigmoid'))
model.add(Dense(units=4, activation='sigmoid'))
model.add(Dense(units=4, activation='sigmoid'))
#model.add(Dense(units=2, activation='sigmoid'))
model.add(Dense(units=1, activation='sigmoid'))
# loss function and optimization
```

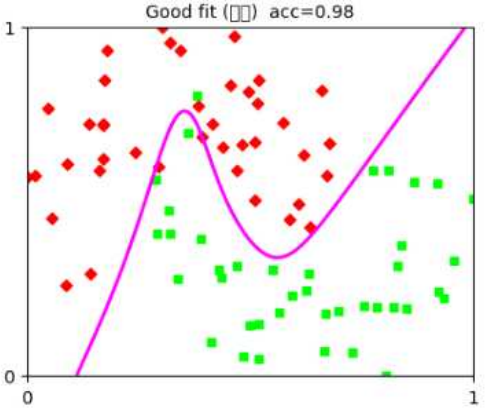
```
/usr/local/lib/python3.13/dist-packages/keras/src/layers/core/dense.p
  super().__init__(activity_regularizer=activity_regularizer, **kwarg
```



중간 모델

```
model = Sequential()
# first dense layer
model.add(Dense(units=2, input_shape=(2,), activation='sigmoid'))
model.add(Dense(units=10, activation='sigmoid'))
model.add(Dense(units=10, activation='sigmoid'))
#model.add(Dense(units=2, activation='sigmoid'))
model.add(Dense(units=1, activation='sigmoid'))
# loss function and optimization
```

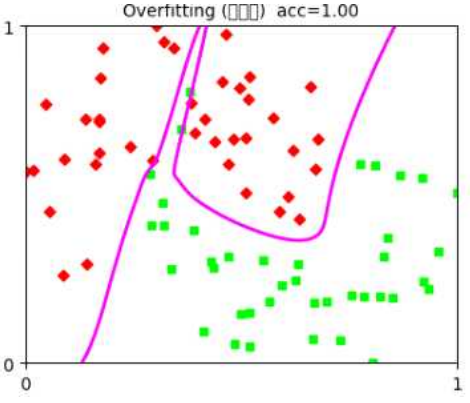
- /usr/local/lib/python3.13/dist-packages/keras/src/layers/core/dense.py



높은 모델

```
model = Sequential()
# first dense layer
model.add(Dense(units=2, input_shape=(2,), activation='sigmoid'))
model.add(Dense(units=8, activation='sigmoid'))
model.add(Dense(units=8, activation='sigmoid'))
#model.add(Dense(units=2, activation='sigmoid'))
model.add(Dense(units=1, activation='sigmoid'))
# loss function and optimization
```

```
/usr/local/lib/python3.13/dist-packages/keras/src/layers/core/dense.
  super().__init__(activity_regularizer=activity_regularizer, **kwar
```



결론

- 모델이 너무 크고 오래 학습하면 노이즈까지 외워 경계가 구불구불해지고, 학습 정확도는 높아도 새 데이터에서는 성능이 떨어진다.(과적합)